Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and affine subschemes

Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.
Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum $\star$ **CORE**

Assume that $\Bbbk$ is algebraically closed and let $X = \operatorname{Spec}(\Bbbk[t])$.

(i) List the points of X.

(ii) Determine which points are closed and compute the closure of each point.

(iii) For a non-zero polynomial f, describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers $\star$ **CORE**

Describe $\operatorname{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically $\star$ **CORE**

Let

$$A = \Bbbk[\varepsilon]/(\varepsilon^2), \qquad X = \operatorname{Spec}(A).$$

Determine the points and open subsets of X. Compare the underlying topological space with $\operatorname{Spec}(\Bbbk)$, and explain why the two affine schemes are nevertheless different.

Hint 1 (conceptual). Every prime ideal contains every nilpotent element. Apply this to the class of ε.

Hint 2 (conceptual). The topology only sees prime ideals. To distinguish the two schemes, compare their rings of global sections and ask whether these rings are reduced.

Exercise 4. A quotient map on spectra $\star$ **CHOICE**

Consider the quotient homomorphism

$$\Bbbk[t] \longrightarrow \Bbbk[t]/(t^2).$$

Describe the induced map on spectra and its image. Why does the image depend only on $\sqrt{(t^2)}$?

Hint 1 (conceptual). For a quotient $A \to A/I$, the induced map identifies $\operatorname{Spec}(A/I)$ with the closed subset $V(I)$. Prime ideals contain I exactly when they contain $\sqrt{I}$.

Exercise 5. Closed subsets in the node $\star\star$ **CHOICE**

Let $A = \Bbbk[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$. Denote the classes of x, y by the same letters.

(i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.

(ii) Compute the closures of the points (x) and (y).

(iii) Show that neither of these two points is closed.

Hint 1 (conceptual). A prime ideal containing xy contains x or y. Also recall that the closure of the point $\mathfrak{p}$ in $\operatorname{Spec}(A)$ is $V(\mathfrak{p})$.

Hint 2 (technical). A point $\mathfrak{p}$ is closed precisely when $\mathfrak{p}$ is maximal. Compare (x) and (y) with the maximal ideal (x, y).

Exercise 6. The affine line over a function field $\star\star$ **CHOICE**

Let $K = \mathbb{C}(t)$ and let $X = \operatorname{Spec}(K[x])$.

(i) Describe the points and the closed points of X.

(ii) Which closed points are K-rational? Give a closed point which is not K-rational.

(iii) Put $S = \mathbb{C}[t] \setminus \{0\}$. Use $K[x] = S^{-1}\mathbb{C}[t,x]$ to identify X with the prime ideals of $\mathbb{C}[t,x]$ lying over $(0) \in \operatorname{Spec}(\mathbb{C}[t])$.

Suggested strategy. Treat $K[x]$ exactly as a polynomial ring over an arbitrary field, then use the prime correspondence for the localisation $S^{-1}\mathbb{C}[t,x]$.

Hint 1 (technical). The non-zero prime ideals of the PID $K[x]$ are generated by irreducible polynomials. Such a point is K-rational exactly when its residue field is K.

Hint 2 (technical). To produce a non-rational point, find an irreducible polynomial of degree 2 in $K[x]$; for example, decide whether t is a square in $\mathbb{C}(t)$.

Hint 3 (technical). A prime $\mathfrak{q} \subset \mathbb{C}[t,x]$ survives the localisation precisely when $\mathfrak{q} \cap S = \varnothing$. Rewrite this as a condition on $\mathfrak{q} \cap \mathbb{C}[t]$.

Thursday afternoon — material available: Lectures 6–9

Exercise 7. Principal opens in the node ⋆⋆ **CORE**

Let $A = \Bbbk[x,y]/(xy)$ and $X = \operatorname{Spec}(A)$.

(i) Describe $D(x)$ and $D(y)$ as affine schemes.

(ii) Show that $D(x) \cap D(y) = \varnothing$ and identify the point missing from $D(x) \cup D(y)$.

(iii) Compute the residue fields at (x), (y) and (x,y).

Hint 1 (technical). Compute A_x and A_y first. The equation $xy = 0$ simplifies drastically after either x or y is inverted.

Hint 2 (technical). Use $D(x) \cap D(y) = D(xy)$ and remember that $xy = 0$ in A.

Hint 3 (technical). For a prime $\mathfrak{p}$, use

$$\kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}).$$

Apply this separately to (x), (y) and (x,y).

Exercise 8. Sections and a stalk on the affine line ⋆ **CORE**

Let $X = \operatorname{Spec}(\Bbbk[t])$ and $U = D(t(t-1))$.

(i) Compute $\mathcal{O}_X(U)$.

(ii) Compute the stalk $\mathcal{O}_{X,(t-a)}$ and its residue field.

(iii) For which a does $(t-a)$ belong to U?

Hint 1 (technical). For an affine scheme, $\Gamma(D(f), \mathcal{O}) = A_f$. Here $f = t(t-1)$.

Hint 2 (technical). The point $(t-a)$ belongs to $D(f)$ if and only if $f \notin (t-a)$, equivalently if and only if $f(a) \neq 0$.

Exercise 9. A subscheme of the affine line with two support points ⋆⋆ **CORE**

Let

$$Z = \operatorname{Spec}\bigl(\Bbbk[t]/(t^3(t-1)^2)\bigr).$$

(i) Determine the underlying topological space of Z.

(ii) Decompose its coordinate ring as a product of two rings.

(iii) Compare the two local pieces with the reduced points at 0 and 1.

Suggested strategy. First take the radical to find the support. Then separate the two coprime primary factors by the Chinese remainder theorem.

Hint 1 (technical). The ideals (t^3) and $((t-1)^2)$ are comaximal. Apply the Chinese remainder theorem before interpreting the two factors geometrically.

Hint 2 (conceptual). Each factor has the topology of one reduced point, but its maximal ideal contains nilpotents. Compare their vector-space lengths.

Exercise 10. Scheme-theoretic unions and intersections ⋆ **CORE**

In $\mathbb{A}^2_{\Bbbk}$, let
$$X = \mathbb{V}(x), \qquad Y = \mathbb{V}(y), \qquad Y' = \mathbb{V}(y^2).$$
Compute the ideals defining $X \cup Y$, $X \cap Y$, $X \cup Y'$ and $X \cap Y'$. Compare their underlying closed subsets.

Hint 1 (conceptual). For closed subschemes of an affine scheme, union corresponds to intersection of ideals and intersection corresponds to sum of ideals.

Hint 2 (technical). Compute $(x) \cap (y)$ and $(x) \cap (y^2)$ using divisibility in the UFD $\Bbbk[x, y]$. Take radicals only after finding the scheme-theoretic ideals.

Exercise 11. Products of rings and disjoint unions $\star\star$ **CHOICE**

Let B and C be non-zero rings.

(i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.

(ii) Deduce that
$$\operatorname{Spec}(B \times C) \simeq \operatorname{Spec}(B) \amalg \operatorname{Spec}(C).$$

(iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

Hint 1 (conceptual). Use the complementary idempotents $e = (1, 0)$ and $1 - e = (0, 1)$. Since $e(1 - e) = 0$, a prime ideal must contain one of them.

Hint 2 (technical). If $(1, 0)$ lies in a prime P, then $B \times 0 \subset P$. Identify the remaining part of P with a prime ideal of C, and argue symmetrically in the other case.

Hint 3 (conceptual). The opens $D(1, 0)$ and $D(0, 1)$ are disjoint, open and closed, and cover the spectrum.

After Friday's arithmetic lecture or homework

Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$ $\star\star$ **CHOICE**

Let
$$A = \mathbb{Z}[s]/(s^2 - 3), \qquad f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(\mathbb{Z}).$$
For a prime p, describe the points of $\operatorname{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbf{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Suggested strategy. Compute the fibre ring over (p) and read its points from the irreducible factorisation of $s^2 - 3$ in $\mathbf{F}_p[s]$.

Hint 1 (technical). Treat $p = 2$ and $p = 3$ directly: the polynomial has a repeated root in each case.

Hint 2 (technical). For $p \neq 2, 3$, distinguish whether 3 is a quadratic residue modulo p. In the irreducible case, determine the residue field of the unique point.

Exercise 13. An arithmetic conic $\star\star\star$ **OPTIONAL**

For
$$X = \operatorname{Spec}\big(\mathbb{Z}[x, y]/(x^2 + y^2 - 5)\big) \longrightarrow \operatorname{Spec}(\mathbb{Z}),$$
write down the equation obtained after reduction modulo a prime p. Determine for which p that equation is a square or has a singular point.

Suggested strategy. Analyse the fibre over each prime by reducing the equation modulo p. For odd p, use the partial derivatives to locate singular points; handle characteristic 2 separately.

Hint 1 (technical). For odd p, a singular point must satisfy $2x = 2y = 0$ as well as the equation. Substitute the resulting candidate into $x^2 + y^2 - 5$.

Hint 2 (technical). In characteristic 2, use $(x + y + 1)^2 = x^2 + y^2 + 1$. At $p = 5$, decide whether -1 is a square and factor the quadratic form.